Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: Data Cleaning and Transformation using Power Query

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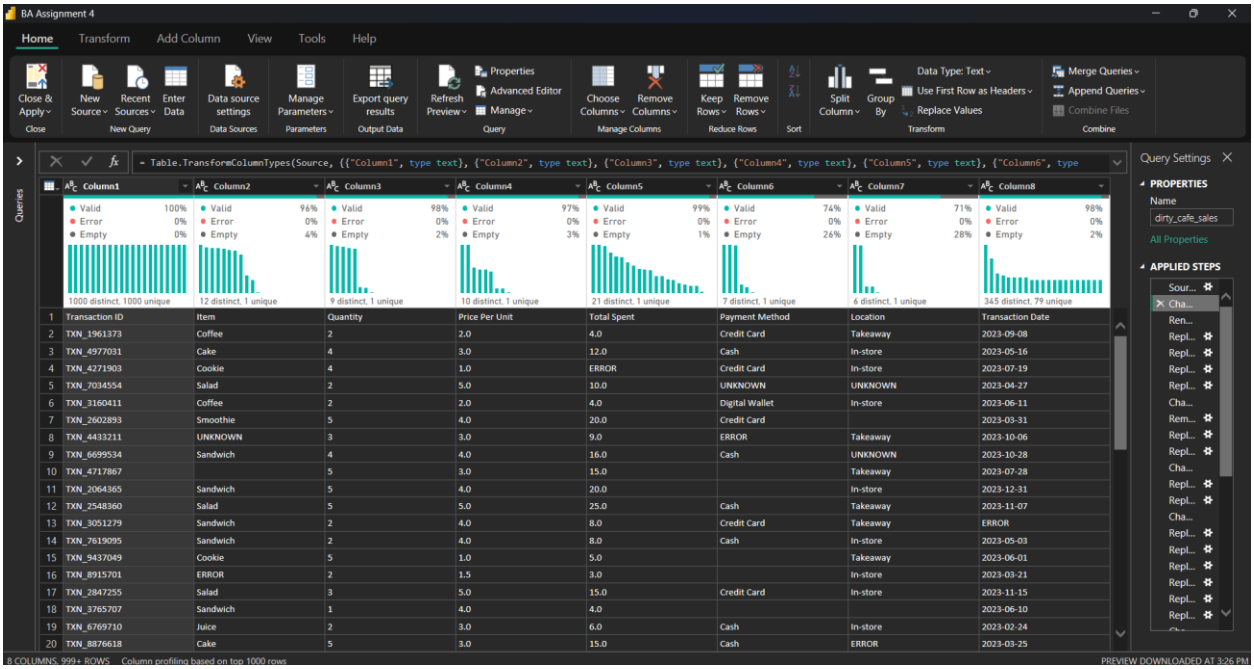
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Data Cleaning and Transformation using Power Query

1. Introduction
This assignment demonstrates the cleaning and preparation of a dirty café sales dataset using Microsoft Power Query. The dataset contained missing values, UNKNOWN/ERROR values, incorrect or inconsistent data types, incomplete records and duplicate transactions. Power Query was used to transform the raw data into a clean and analysis-ready table.

2. Objectives
- Import and inspect the dirty café sales dataset.
- Identify missing, blank, UNKNOWN and ERROR values.
- Set appropriate data types for text, numeric and date columns.
- Clean categorical fields such as Item, Payment Method and Location.
- Create cleaned columns for Quantity, Price Per Unit and Total Spent.
- Remove incomplete rows and duplicate transactions.
- Remove the original columns after creating cleaned versions.
- Rename the cleaned columns back to meaningful final names.
- Produce a final clean dataset suitable for analysis.

3. Dataset Overview
The source table is named “dirty_cafe_sales”. The raw dataset contains café transaction information including Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location and Transaction Date.



	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
1	TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway	2023-09-08
2	TXN_4977031	Cake	4	3.0	12.0	Cash	In-store	2023-05-16
3	TXN_4271903	Cookie	4	1.0	ERROR	Credit Card	In-store	2023-07-19
4	TXN_7034554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN	2023-04-27
5	TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store	2023-06-11
6	TXN_2602893	Smoothie	5	4.0	20.0	Credit Card		2023-03-31
7	TXN_4433211	UNKNOWN	3	3.0	9.0	ERROR	Takeaway	2023-10-06
8	TXN_6699534	Sandwich	4	4.0	16.0	Cash	UNKNOWN	2023-10-28
9	TXN_4717867		5	3.0	15.0		Takeaway	2023-07-28
10	TXN_2064365	Sandwich	5	4.0	20.0		In-store	2023-12-31
11	TXN_2548360	Salad	5	5.0	25.0	Cash	Takeaway	2023-11-07
12	TXN_3051279	Sandwich	2	4.0	8.0	Credit Card	Takeaway	ERROR
13	TXN_7619095	Sandwich	2	4.0	8.0	Cash	In-store	2023-05-03
14	TXN_9437049	Cookie	5	1.0	5.0		Takeaway	2023-06-01
15	TXN_8915701	ERROR	2	1.5	3.0		In-store	2023-03-21
16	TXN_2847255	Salad	3	5.0	15.0	Credit Card	In-store	2023-11-15
17	TXN_3765707	Sandwich	1	4.0	4.0			2023-06-10
18	TXN_6769710	Juice	2	3.0	6.0	Cash	In-store	2023-02-24
19	TXN_8876618	Cake	5	3.0	15.0	Cash	ERROR	2023-03-25

Figure 1: Raw dataset in Power Query before cleaning.

4. Data Cleaning and Transformation Process

4.1 Initial Data Types and Column Names

The initial data types were assigned and the original columns were renamed where required. This established a consistent structure before further transformations.

4.2 Cleaning Invalid and Unknown Values

UNKNOWN and ERROR values were replaced or converted into usable values. Separate replacement steps were applied to Item, Quantity, Price Per Unit, Total Spent, Payment Method and Location.

4.3 Removing the Extra Header Row

An unwanted top row was removed using the Skip Rows transformation so that the actual transaction records could be processed correctly.

4.4 Numeric Data Cleaning

Quantity, Price Per Unit and Total Spent were cleaned. Price Per Unit was converted to a decimal number. Where one numeric field was missing but the other values were available, custom columns were created to calculate or recover the missing value.

4.5 Date Data Type

Transaction Date was converted to the Date data type so that it could be handled correctly as a date field.

4.6 Creation of Clean Columns

Five clean columns were created: Payment Method Clean, Location Clean, Item Clean, Quantity Clean, Price Per Unit Clean and Total Spent Clean. Missing categorical values were represented as “Not Specified”.

4.7 Removing Incomplete Rows

Rows with missing essential transaction information were filtered out. This ensured that the final dataset contained complete records for the required fields.

4.8 Removing Duplicate Transactions

Duplicate transaction records were removed using the Remove Duplicates transformation.

4.9 Removing Original Columns

After the cleaned columns were created and verified, the original unclean columns were removed.

4.10 Renaming Clean Columns

The cleaned columns were renamed to the final business-friendly names: Item, Quantity, Price Per Unit, Total Spent, Payment Method and Location.

4.11 Final Data Types and Blank Value Replacement

Final data types were applied to the cleaned columns. Remaining blank Item and Location values were replaced with “Not Specified”.

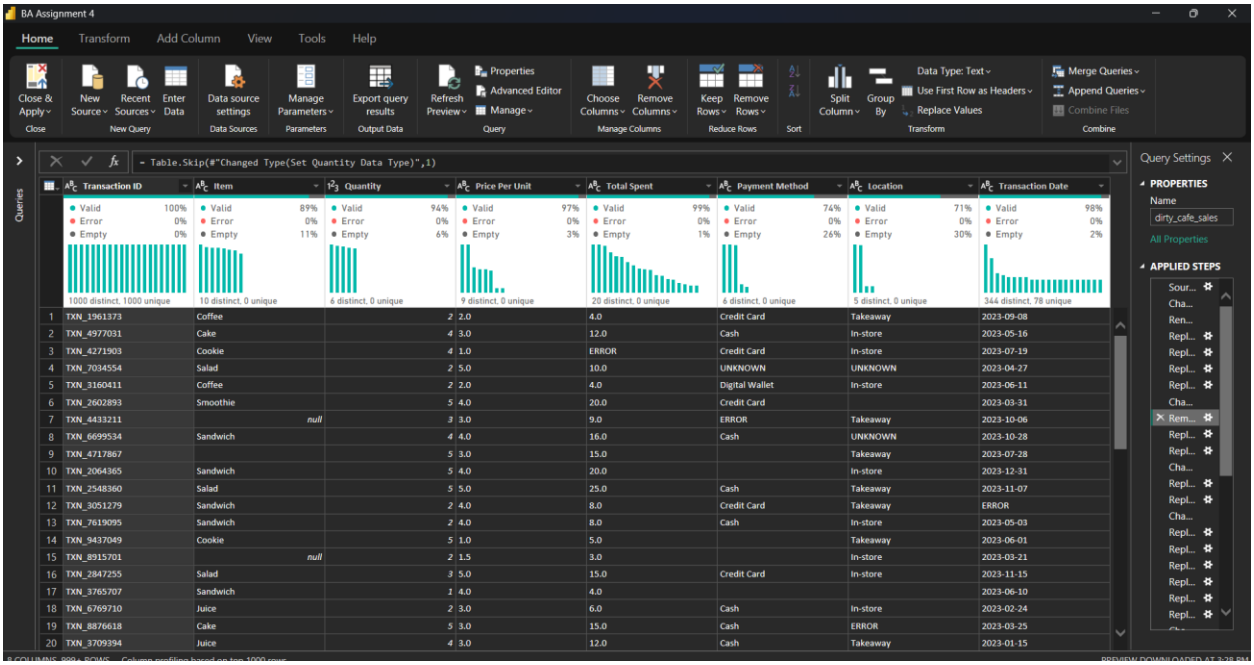


Figure 2: Removing the unwanted top row using Skip Rows.

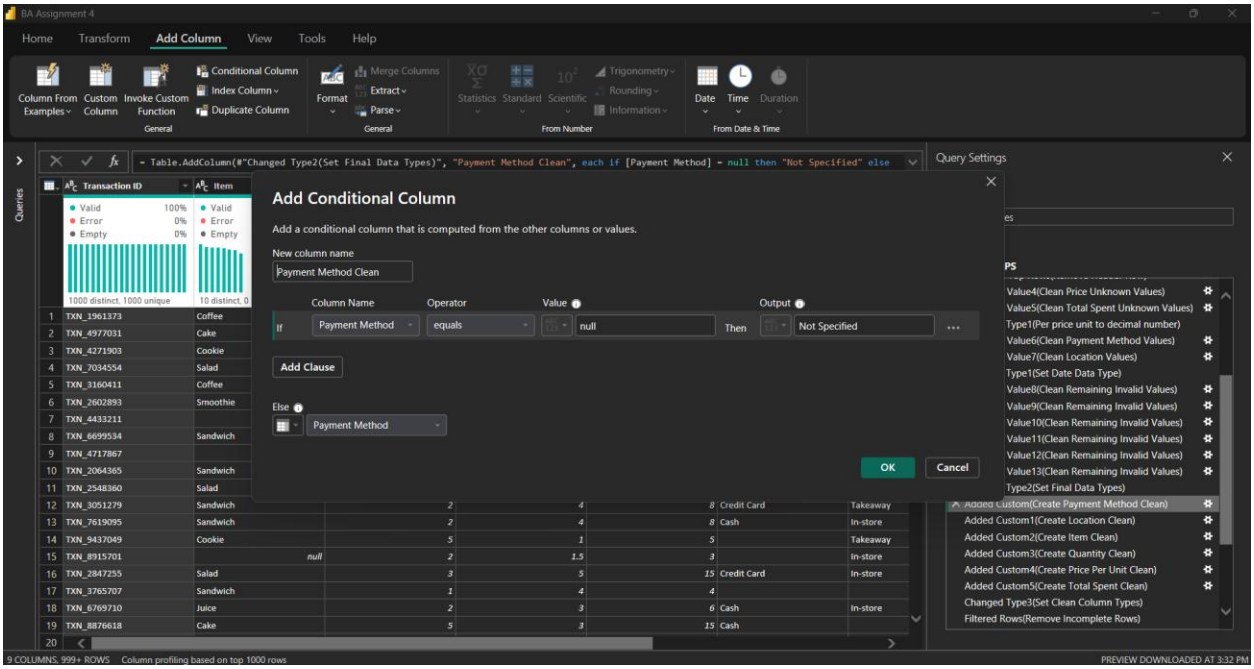


Figure 3: Creating Payment Method Clean using a conditional column.

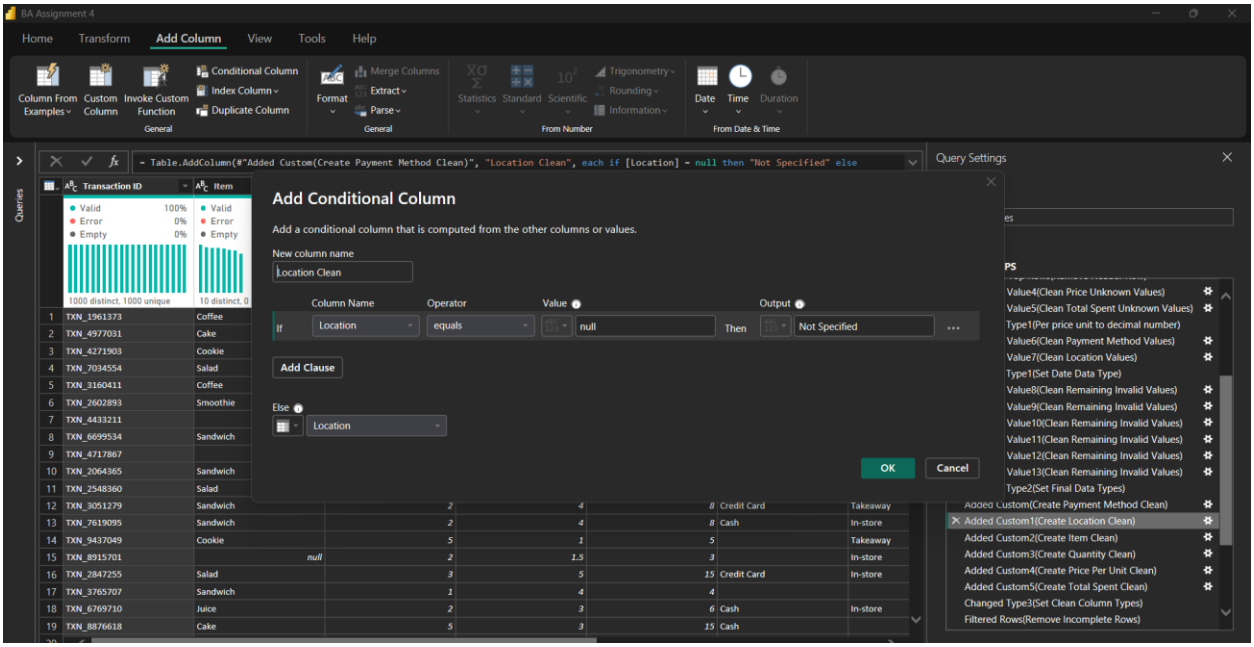


Figure 4: Creating Location Clean using a conditional column.

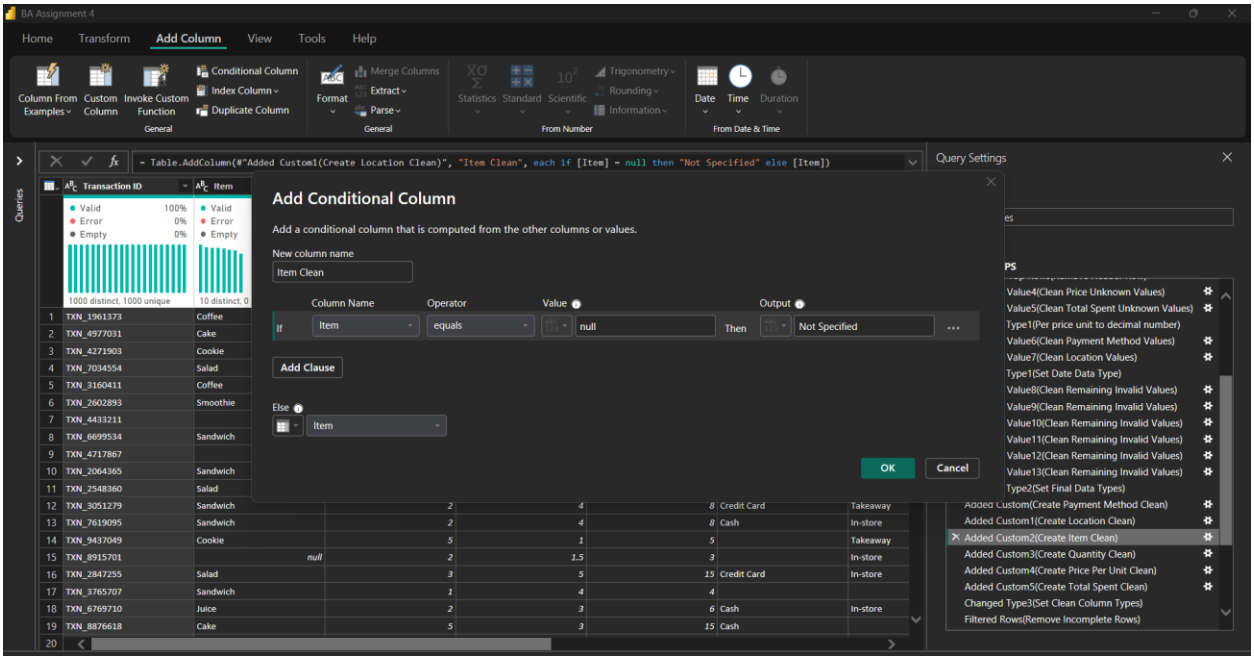


Figure 5: Creating Item Clean using a conditional column.

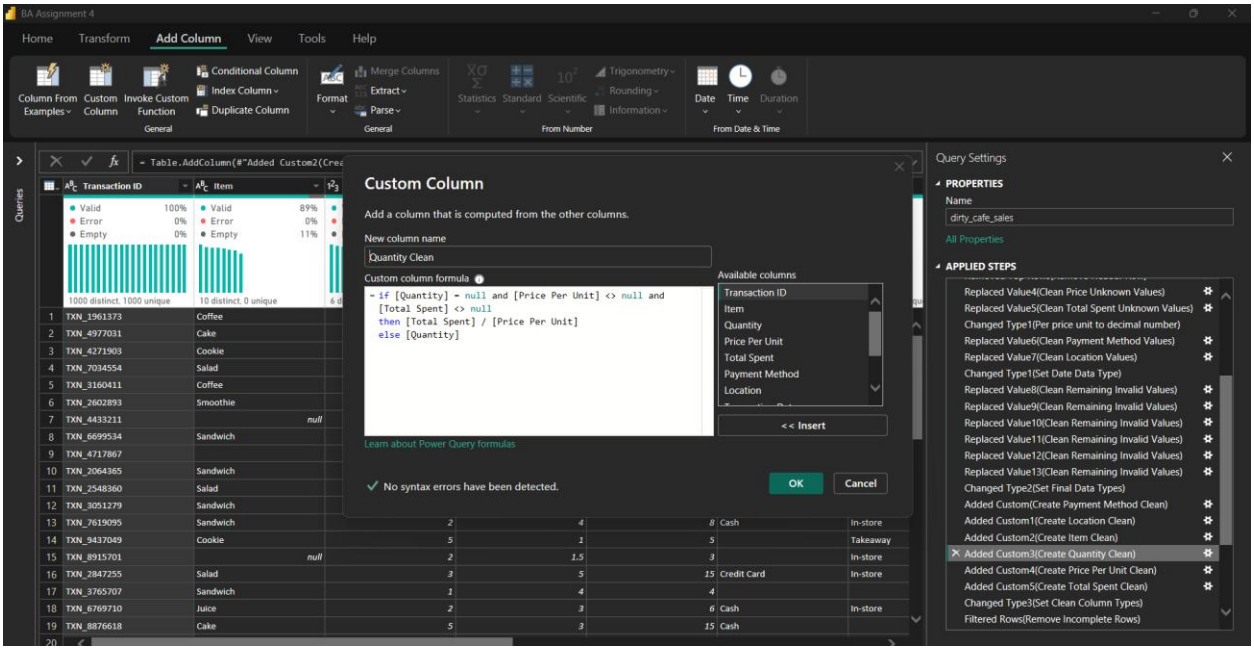


Figure 6: Creating Quantity Clean using a custom formula.

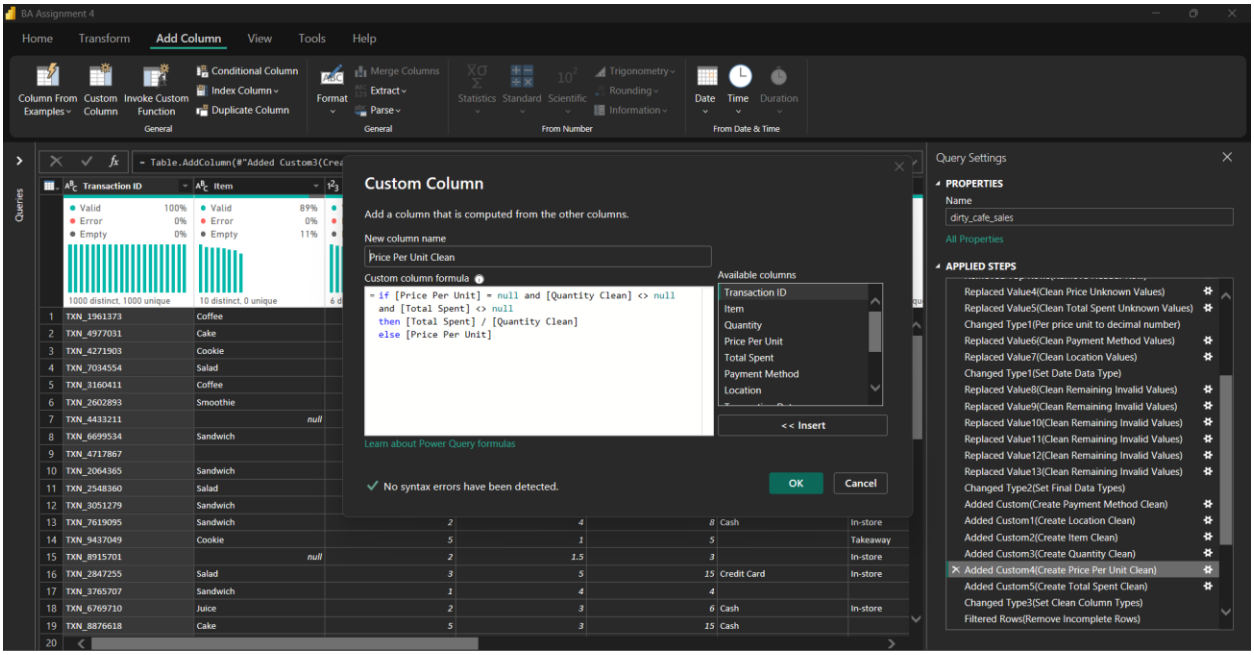


Figure 7: Creating Price Per Unit Clean using a custom formula.

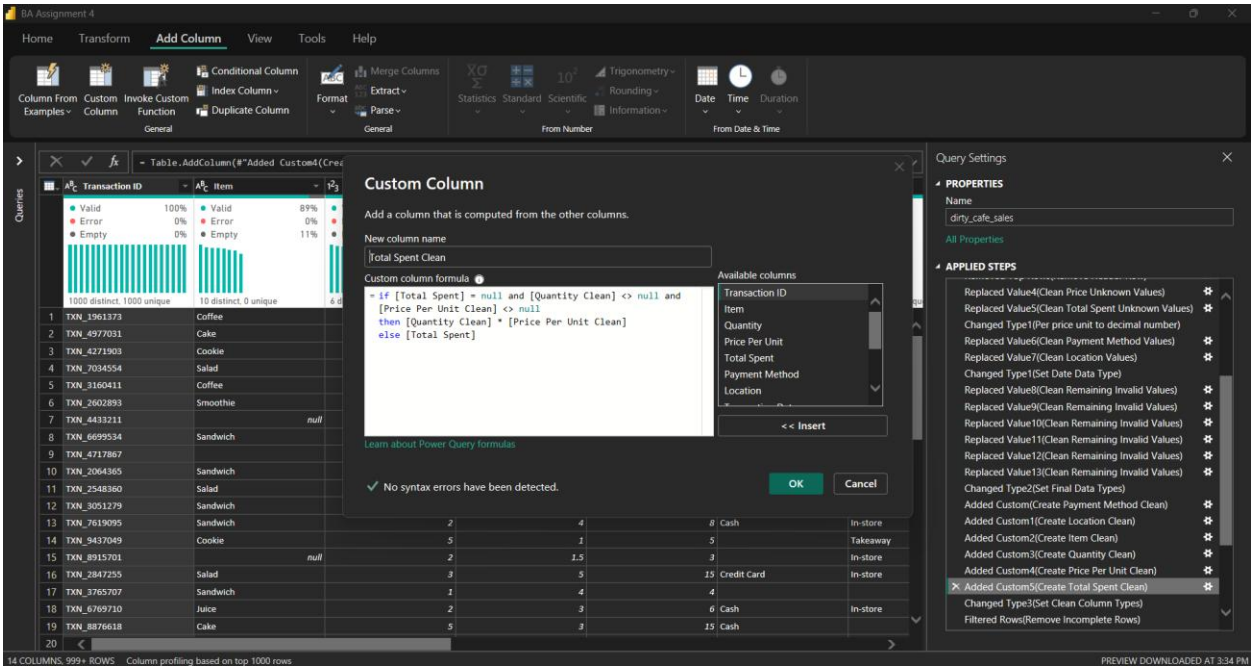


Figure 8: Creating Total Spent Clean using a custom formula.

5. Applied Steps in Power Query

The Query Settings pane records the transformations performed on the dataset. The applied steps provide a reproducible sequence of the data-cleaning process.

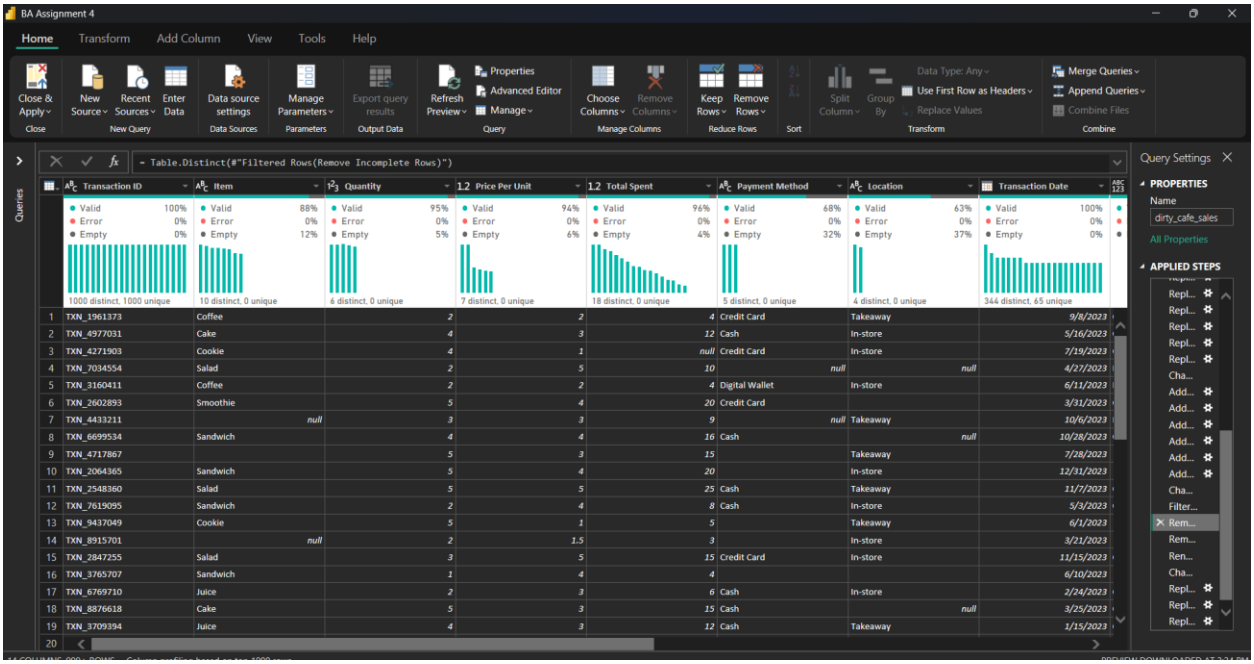


Figure 9: Applied Steps showing the complete transformation workflow.

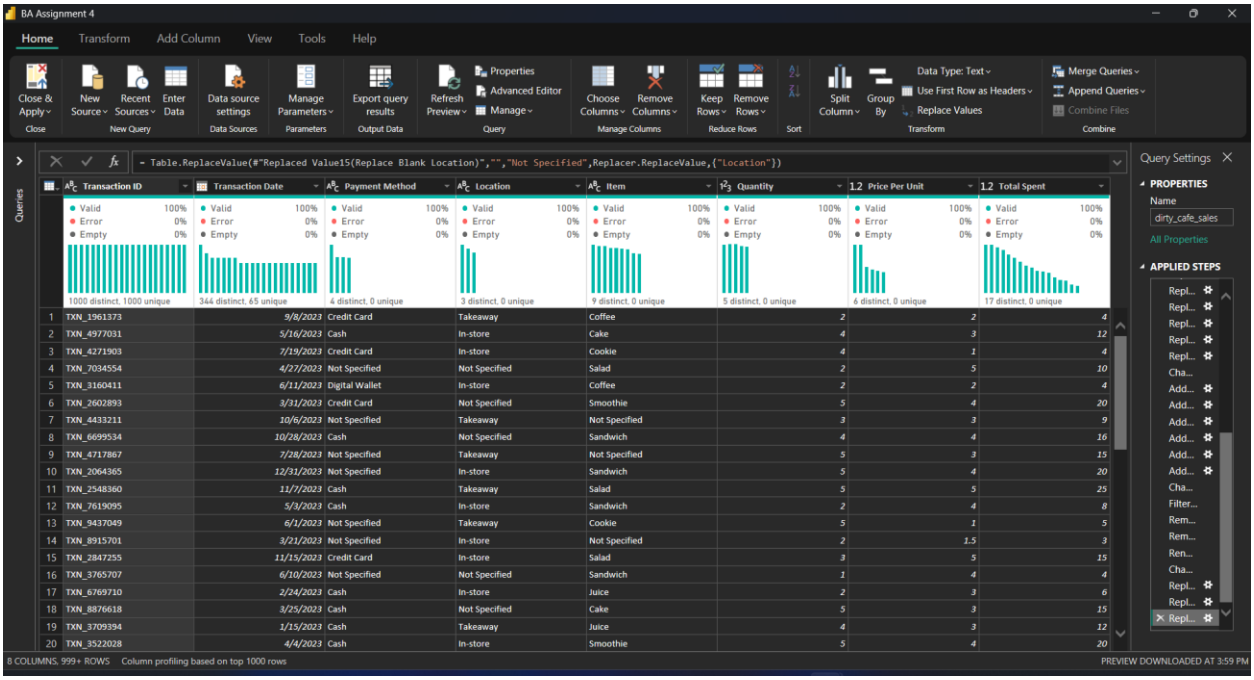


Figure 10: Lower portion of Applied Steps showing clean columns, filtering, duplicate removal and final renaming.

6. Final Cleaned Dataset

After completing the transformations, the final table contains the cleaned transaction fields. The cleaned categorical fields use “Not Specified” where appropriate, numeric fields have usable values, and the dataset has been filtered to remove incomplete and duplicate records.

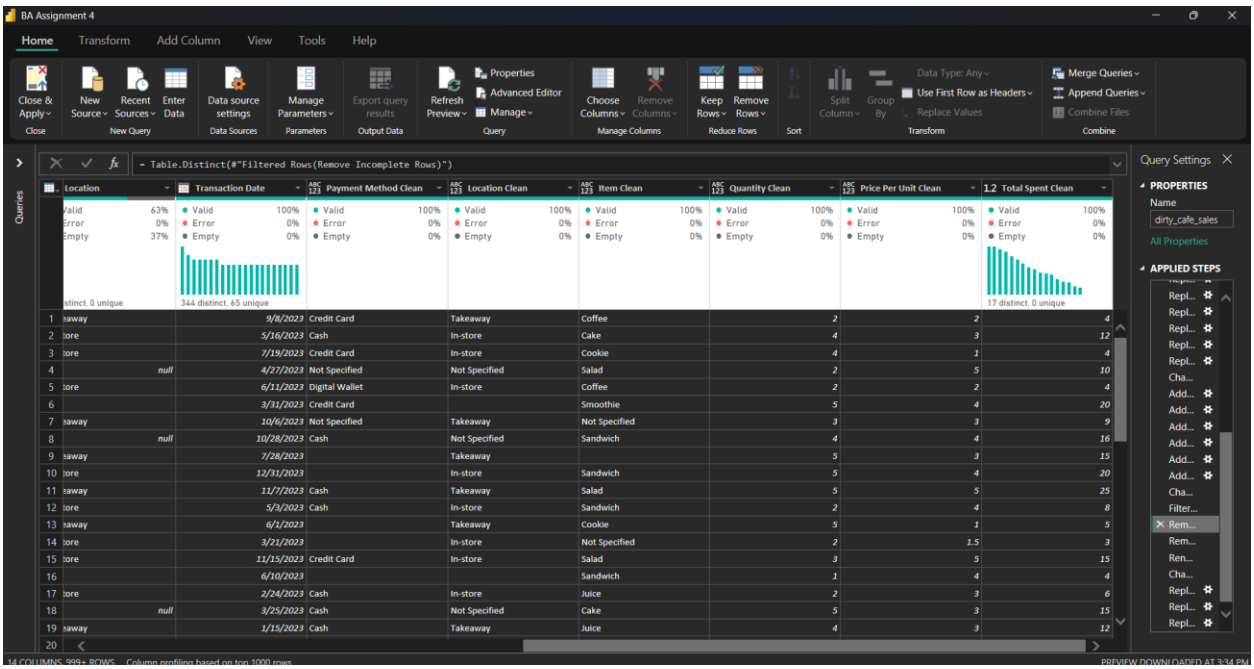


Figure 11: Final cleaned café sales dataset after all transformations.

7. Additional Evidence – Applied Steps

The following screenshots provide additional evidence of the Power Query Applied Steps. They show the sequence of transformations used to clean, create, filter, remove duplicates, remove original columns, rename the cleaned columns, set final data types, and replace remaining blanks.

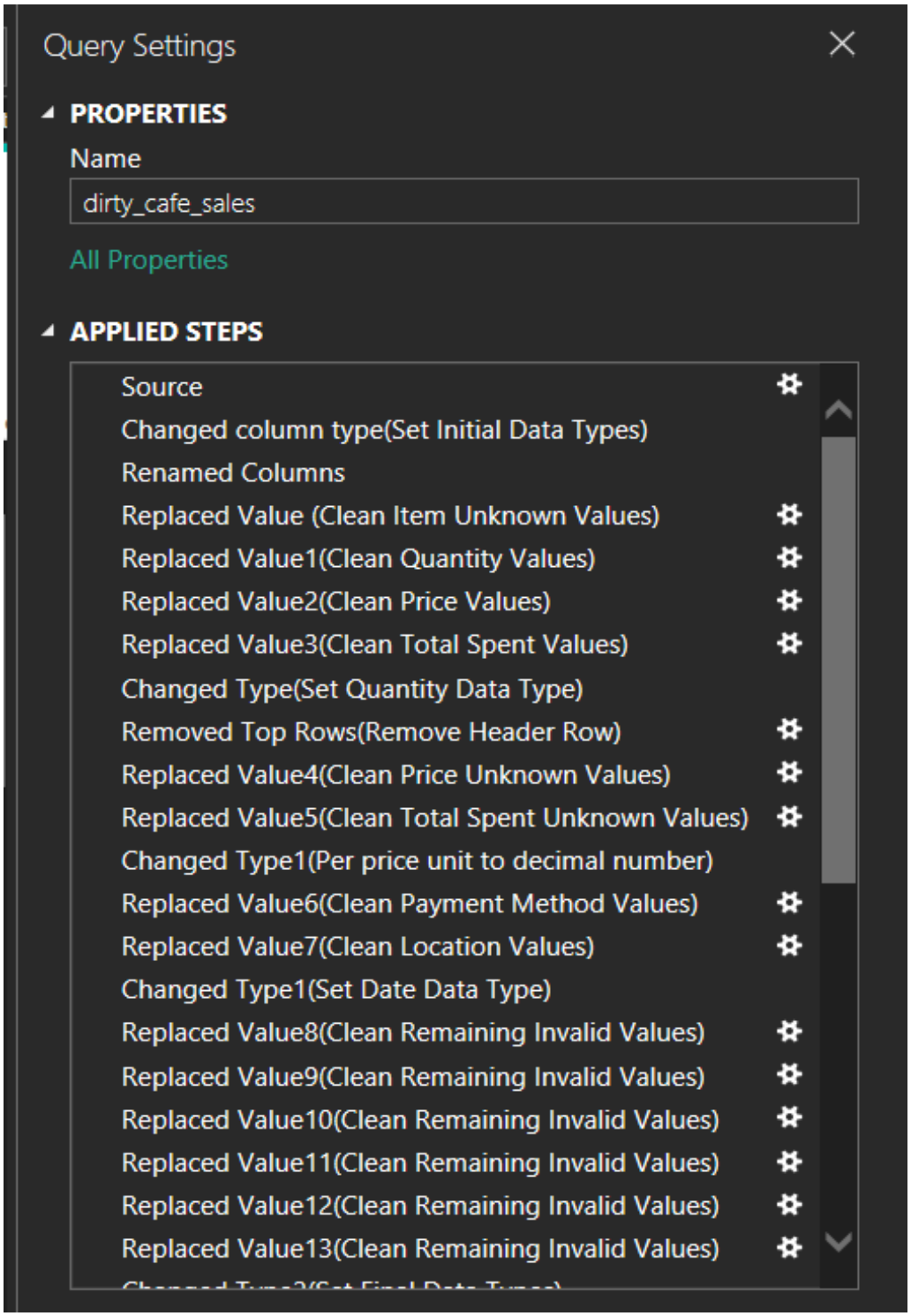


Figure 12: Applied Steps – initial data preparation and cleaning transformations.

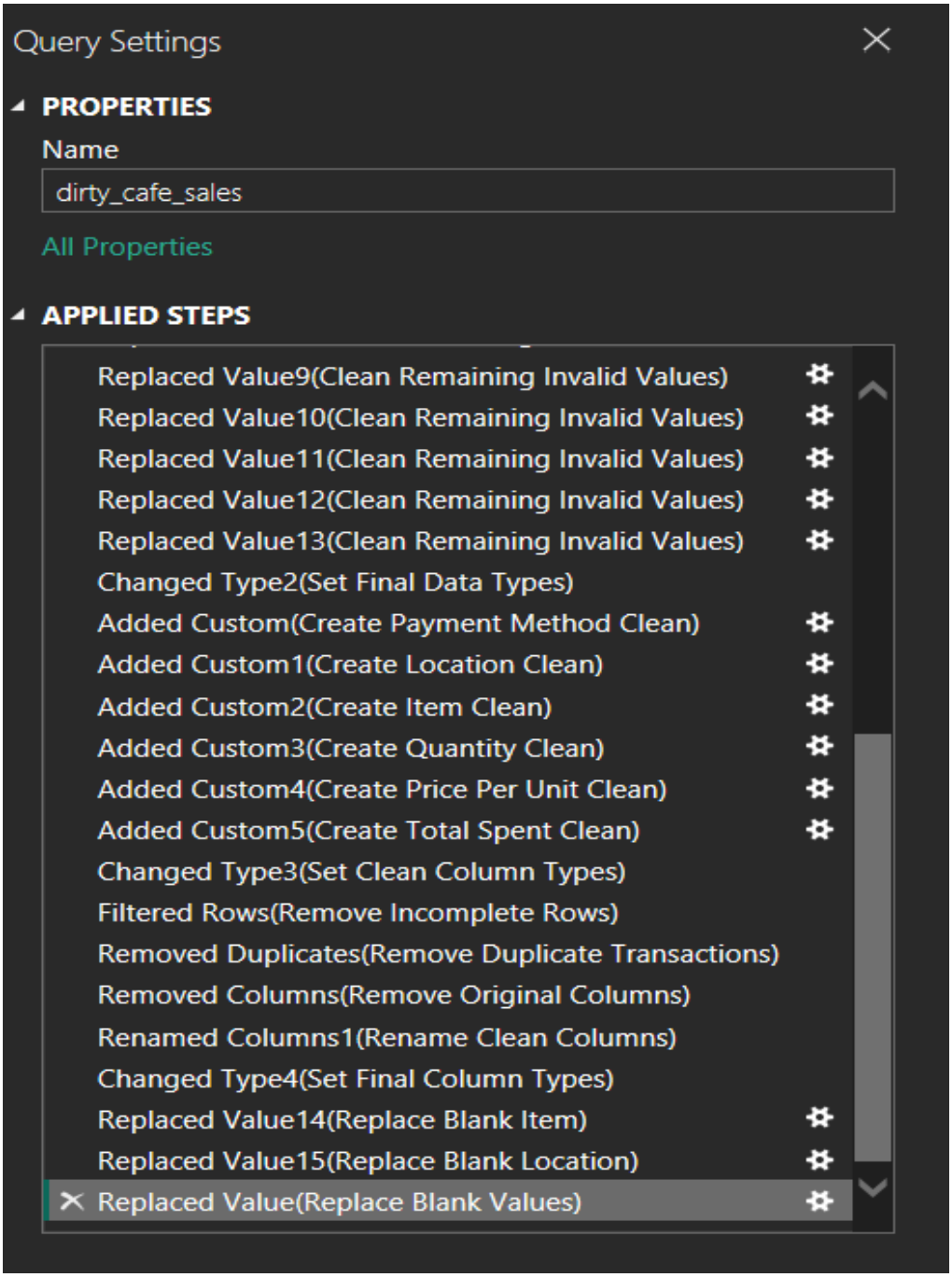


Figure 13: Applied Steps – creation of clean columns, filtering, duplicate removal, renaming, final data types and blank-value replacement.

8. Result

The dirty café sales data was successfully transformed into a cleaner and more consistent dataset. The final query contains business-friendly column names and standardized values, making the data ready for further Business Analytics tasks.

Area	Result
Missing categorical values	Handled using “Not Specified”
Invalid / UNKNOWN / ERROR values	Cleaned through replacement transformations
Numeric fields	Cleaned and assigned suitable numeric types
Transaction Date	Converted to Date type
Incomplete records	Filtered out
Duplicate records	Removed
Final columns	Renamed to business-friendly names
Final dataset	Clean and analysis-ready

9. Conclusion

Power Query provided an efficient and repeatable method for cleaning the café sales dataset. The transformation process addressed data-quality problems step by step, including invalid values, missing values, data-type issues, incomplete rows and duplicate transactions. The resulting dataset is more reliable and suitable for subsequent Business Analytics analysis.